

Toward Principled Threshold Selection for Blink Region Detection in Biological Signals: A Cross-Dataset Benchmark with LLM-Guided Candidate Derivation

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Abstract

Blink detection from electroencephalography (EEG) serves two complementary purposes: blinks are a major physiological contaminant that must be identified for artifact handling, and they are also informative behavioral markers linked to arousal, fatigue, cognitive load, and operational performance. Despite substantial progress in blink detection, a practically important setting remains unaddressed: epoch-structured recordings in which a long continuous signal is divided into non-overlapping analysis windows and some windows are pre-rejected by the analyst before automated processing begins. This setting is the norm in driving neuroscience and other neuroergonomic paradigms, yet existing detectors—including BLINKER and MNE-based amplitude routines—operate on concatenated signals and implicitly pool blink-containing and blink-free epochs when estimating thresholds. We show that this pooling systematically biases the threshold upward and reduces recall.

We propose a three-stage epoch-aware pipeline. Stage A screens epochs by their peak-to-peak amplitude via an autoreject-inspired cross-validation criterion, yielding a set of suspicious (likely blink-containing) epochs. Stage B estimates a robust sample-level threshold from the suspicious epoch pool using the median and median absolute deviation (MAD), concentrating estimation on signal segments enriched for blink-like excursions. Stage C applies the threshold to extract blink regions from the suspicious epochs and maps detections back to epoch-local coordinates. A fallback mechanism ensures stability when too few suspicious epochs are available.

We evaluate five conditions—BLINKER with naive concatenation, direct Bayesian threshold optimisation, MNE amplitude annotation, and the proposed pipeline with mean-based and median-based estimators—on a large naturalistic driving corpus under epoch durations of 20, 30, 60, and 120 seconds. Statistical comparisons use macro-averaged event-level F1 with Wilcoxon signed-rank tests. Additional analyses examine sensitivity to the minimum suspicious-epoch count $n_{\min}$, stability across boundary tolerances, robustness under user-initiated epoch pre-rejection, and morphological characteristics of missed and spurious blinks. Results demonstrate that epoch-aware threshold estimation consistently outperforms naive concatenation, that the robust MAD-based estimator is preferable to a mean-based alternative, and that the pipeline degrades gracefully as the available epoch pool shrinks due to pre-rejection.

1 Introduction

Electroencephalography is routinely affected by ocular activity, and blink contamination remains one of the oldest and most persistent obstacles in EEG analysis. At the same time, spontaneous and task-related blinks are themselves scientifically meaningful: prior work links blink-derived measures to fatigue, drowsiness, attention, cognitive load, and operational performance, and more recent blink-linked EEG studies show that blink-locked neural activity can index cognitive demand in naturalistic tasks (Croft and Barry, 2000; Kleifges et al., 2017; Alyan et al., 2023). These dual roles make blink detection a foundational problem rather than a preprocessing detail: researchers

need detectors accurate enough for artifact handling, and reliable enough to preserve blink events as analyzable signals in their own right.

Thresholding remains central to this problem because even sophisticated pipelines frequently begin with amplitude gates. Threshold-based designs are attractive in practice because they are interpretable, fast, and deployable in few-channel and real-time environments—properties that are especially important for frontal wearable EEG and neuroergonomic settings, where channel count, latency, and computational budget are constrained. However, thresholding is also where many detectors fail: a threshold learned from background-dominated samples may drift toward over-detection, while a threshold tuned to one subject or experiment can collapse under inter-subject variability or under distributions that include eye movements, muscle activity, or epileptiform events (Chang et al., 2016; Kleifges et al., 2017; Agarwal and Sivakumar, 2019; Tran et al., 2021; M. Wang et al., 2022; Y. Zhang et al., 2023).

The epoch-structured setting. A critical and largely overlooked challenge arises in domain-specific EEG paradigms where analysis is inherently epoch-based. In driving research, for example, a continuous 60–90 minute recording is routinely segmented into non-overlapping 20–120 second epochs that correspond to task events, environmental changes, or regularly spaced analysis windows (Z. Cao et al., 2019). Blink detection in this context means detecting blinks within the epoch structure, not within an undivided recording. Furthermore, in practice not all epochs are available for analysis: some are pre-rejected by the analyst because of gross artefacts, electrode failures, or brief signal loss, leaving a subset of valid epochs that may be non-contiguous and non-representative of the full recording.

To our knowledge, no existing blink-detection system explicitly addresses this setting. State-of-the-art tools such as BLINKER (Kleifges et al., 2017) and MNE-Python’s amplitude-annotation routines operate on continuous or concatenated signals and derive thresholds from the full recording. When applied naively to an epoch-structured recording—by concatenating available epochs and treating the result as a continuous signal—these tools implicitly pool blink-containing and blink-free epochs when estimating the threshold. This pooling is directly harmful: clean epochs dilute the amplitude statistics that determine where the threshold is placed, systematically biasing the threshold upward and suppressing recall. The problem is compounded when some epochs have been pre-rejected, because the remaining valid epochs may be skewed toward a particular recording condition.

Threshold selection as a first-class problem. Beyond the epoch-structured setting, there is a broader gap in the literature: threshold selection is rarely studied as an isolated problem. The choice between a global mean-plus-standard-deviation rule, a robust median-MAD rule, a locally adaptive estimate, and a data-driven optimised threshold can substantially alter detected event counts and boundary precision, yet these choices are frequently absorbed into larger pipelines and not evaluated independently. Without a systematic comparison under a shared evaluation protocol, it is not possible to attribute performance differences to threshold design versus other pipeline choices.

Contributions. This paper makes four contributions. First, we formalise the epoch-structured blink detection setting and show that naive concatenation of epochs leads to measurably worse performance than epoch-aware threshold estimation. Second, we propose a three-stage pipeline—epoch screening, sample-level threshold estimation, and blink-region extraction—that explicitly accounts for the epoch structure and handles pre-rejected epochs gracefully by design. Third, we introduce

and evaluate a robust threshold estimator at Stage B based on the median and MAD, and compare it systematically against mean-based and directly optimised alternatives. Fourth, we provide a comprehensive benchmark covering five competing methods, four epoch durations, sensitivity analyses for the fallback threshold $n_{\min}$ and the boundary tolerance, and a morphological analysis of detection failures. All experiments are conducted on a large naturalistic driving EEG corpus and evaluated under an event-level overlap criterion with macro-averaged F1 and Wilcoxon signed-rank tests.

2 Related Work

The classical literature established the difficulty of ocular contamination and motivated automated statistical detection directly from EEG. Early work emphasized artifact removal and statistical screening rather than blink-region benchmarking per se, but it already showed that extreme-value behavior, channel context, and morphology matter (Croft and Barry, 2000; Klein and Skrandies, 2013). This historical backdrop explains why simple one-value global thresholds can be attractive, yet brittle.

A first important stream comprises single-channel or few-channel threshold-centered detectors. Chang et al. (2016) detect blink artifacts from a single prefrontal channel using digital filtering and rule-based decisions. Tran et al. (2021) present an explicitly thresholding-based EEG blink detector and compute a minimum amplitude threshold from transformed signal magnitude and signal standard deviation. Y. Zhang et al. (2023) move toward real-time frontal EEG detection with short windows and decision logic around potential blink boundaries. M. Wang et al. (2022) extend thresholding to harder clinical settings by optimizing multidimensional features in the presence of epileptiform discharges, and later work based on higher-order cumulants continues the move toward short-segment single-channel detection (G. Wang et al., 2025).

A second stream contains hybrid thresholding pipelines. BLINKER first identifies candidate intervals with a mean-based standard-deviation threshold and then rejects implausible candidates using robust amplitude-distribution criteria derived from the median absolute deviation (Kleifges et al., 2017). J. Cao et al. (2021) use cascaded thresholding over amplitude, amplitude displacement, and cross-channel correlation before applying a Gaussian mixture model. Agarwal and Sivakumar (2019) self-learn user-specific profiles in an unsupervised manner, while Valderrama et al. (2018) use iterative template matching and automatic threshold/template estimation for suppression in single-channel recordings. Guttmann-Flury, Sheng, D. Zhang, et al. (2019) further emphasize adaptation to inter- and intra-subject variability, underscoring that fixed thresholds are often not enough.

3 Method

3.1 Selection of Bad Epochs

The first stage of the proposed pipeline performs epoch-level screening using an autoreject-inspired threshold selection procedure. Its purpose is not to estimate the onset or offset of a blink, but to identify epochs whose amplitudes are sufficiently abnormal to warrant subsequent sample-level localization. Let $X \in \mathbb{R}^{N \times C \times T}$ denote the epoched multichannel signal after preprocessing, where N is the number of valid epochs, C is the number of channels, and T is the number of samples per epoch. The sample at epoch $i \in \mathcal{E} = \{1, \dots, N\}$, channel $c \in \mathcal{C} = \{1, \dots, C\}$, and within-epoch time index $t \in \{1, \dots, T\}$ is denoted by $x_{i,c,t}$.

For each epoch and channel, an amplitude-range statistic is computed using the peak-to-peak value

$$a_{i,c} = \max_{1 \leq t \leq T} x_{i,c,t} - \min_{1 \leq t \leq T} x_{i,c,t}.$$

Large values of $a_{i,c}$ indicate that the corresponding epoch contains a high-amplitude excursion in channel c , a pattern consistent with ocular or other transient artifacts. Rather than imposing a fixed rejection level, the pipeline estimates a channel-specific threshold τ_c^* from the data. Candidate thresholds are taken from the empirical range of peak-to-peak amplitudes for that channel, denoted $\mathcal{Q}_c = \{a_{i,c} : i \in \mathcal{E}\}$, and are evaluated by a cross-validation criterion.

Specifically, let $\{(\mathcal{T}_k, \mathcal{V}_k)\}_{k=1}^K$ be cross-validation splits of the epoch indices into training sets $\mathcal{T}_k$ and validation sets $\mathcal{V}_k$. For a candidate threshold $\tau \in \mathcal{Q}_c$, the training epochs retained for channel c in fold k are

$$\mathcal{G}_{k,c}(\tau) = \{i \in \mathcal{T}_k : a_{i,c} \leq \tau\}.$$

These retained epochs define a threshold-dependent training template

$$\bar{x}_{k,c}^{(\tau)}(t) = \frac{1}{|\mathcal{G}_{k,c}(\tau)|} \sum_{i \in \mathcal{G}_{k,c}(\tau)} x_{i,c,t},$$

where candidates yielding an empty retained set are treated as inadmissible. The corresponding validation reference is computed as the pointwise median across the validation epochs,

$$\tilde{x}_{k,c}(t) = \text{median}_{i \in \mathcal{V}_k} x_{i,c,t}.$$

The use of a median validation reference makes the criterion less sensitive to large validation-set artifacts. The validation loss for channel c is then defined as

$$\mathcal{J}_c(\tau) = \frac{1}{K} \sum_{k=1}^K \left[\frac{1}{T} \sum_{t=1}^T \left\{ \bar{x}_{k,c}^{(\tau)}(t) - \tilde{x}_{k,c}(t) \right\}^2 \right]^{1/2}.$$

The selected channel threshold is the candidate that minimizes the average validation discrepancy,

$$\tau_c^* = \arg \min_{\tau \in \mathcal{Q}_c} \mathcal{J}_c(\tau).$$

In practice, this objective may be searched by sequential Bayesian optimization or by a random search over candidate peak-to-peak thresholds; both choices retain the same validation objective and rejection rule.

After the thresholds $\{\tau_c^*\}_{c=1}^C$ have been estimated, each epoch receives a channel-level artifact indicator

$$z_{i,c} = \mathbf{1}\{a_{i,c} > \tau_c^*\},$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. The present pipeline uses the union of the channel-level decisions to form the epoch-level screening mask,

$$b_i = \mathbf{1} \left\{ \sum_{c=1}^C z_{i,c} \geq 1 \right\} = \mathbf{1} \{ \exists c \in \mathcal{C} : a_{i,c} > \tau_c^* \}.$$

The set of suspicious epochs is therefore

$$\mathcal{B} = \{i \in \mathcal{E} : b_i = 1\}.$$

These epochs are interpreted as likely blink-contaminated or artifact-heavy segments. They provide the restricted data subset from which the next stage estimates sample-level blink-region thresholds.

3.2 Computation of Sample-Level Blink-Region Threshold Candidates

The second stage converts the epoch-level artifact decision into a sample-level thresholding problem. The autoreject-style screening in the previous stage identifies a set of suspicious epochs $\mathcal{B}$, but its thresholds τ_c^* are defined on peak-to-peak amplitudes over whole epochs. Such thresholds are therefore appropriate for deciding whether an epoch is abnormal, but they are not directly interpretable as sample-wise blink crossing levels. Stage B estimates a separate channel-specific threshold in the amplitude domain of the time series itself.

Let $\mathcal{S}$ denote the set of epochs used for estimating the sample-level threshold. In the primary analysis, $\mathcal{S} = \mathcal{B}$, so that the estimate is driven by epochs already identified as likely to contain blink-related activity. If too few suspicious epochs are available, the method falls back to the full valid epoch set $\mathcal{E}$ to avoid estimating a threshold from an insufficient sample:

$$\mathcal{S} = \begin{cases} \mathcal{B}, & |\mathcal{B}| \geq n_{\min}, \\ \mathcal{E}, & |\mathcal{B}| < n_{\min}, \end{cases}$$

where $n_{\min}$ is the minimum number of screened epochs required for a stable estimate.

For each channel c , all samples from the selected epochs are pooled into a one-dimensional empirical sample set

$$\mathcal{Y}_c = \{x_{i,c,t} : i \in \mathcal{S}, 1 \leq t \leq T\}.$$

This pooling step marks the methodological transition from epoch-level rejection to sample-level localization: the quantities used in Stage B are no longer peak-to-peak ranges $a_{i,c}$, but the individual signal amplitudes $x_{i,c,t}$ that will later be thresholded to form candidate blink regions.

The sample-level threshold is estimated using a robust location-plus-dispersion form. Let ξ_c be the channel-specific center of the sample distribution. The default choice is the median,

$$\xi_c = \text{median}_{y \in \mathcal{Y}_c} y,$$

although the arithmetic mean may be substituted when a mean-centered threshold is required for comparison. Dispersion is estimated by the median absolute deviation (MAD),

$$\text{MAD}_c = \text{median}_{y \in \mathcal{Y}_c} |y - \text{median}_{u \in \mathcal{Y}_c} u|,$$

and is rescaled to a standard-deviation-compatible quantity by

$$\sigma_c^{\text{rob}} = 1.4826 \text{ MAD}_c.$$

The constant 1.4826 is the usual normal-consistency factor for the MAD. For a sensitivity parameter $k > 0$, the resulting sample-level blink-region threshold candidate for channel c is

$$\theta_c = \xi_c + k\sigma_c^{\text{rob}}.$$

The set

$$\Theta = \{\theta_c : c \in \mathcal{C}\}$$

constitutes the channel-wise threshold candidates passed to the blink-region extraction stage.

This construction deliberately separates the statistical role of the two thresholds in the pipeline. The quantities τ_c^* from Stage A are epoch-rejection thresholds defined on peak-to-peak amplitudes, whereas θ_c is a sample-level amplitude threshold defined on the time-series samples themselves. Estimating θ_c from $\mathcal{S}$ concentrates the calculation on signal segments enriched for blink-like excursions, while the robust center and MAD-scaled dispersion reduce the influence of isolated extreme peaks within those same segments.

3.3 Extraction of Blink Regions

The final stage applies the sample-level thresholds $\Theta = \{\theta_c\}_{c=1}^C$ to obtain channel-wise blink regions. For each channel c , the epochs selected for localization are ordered as $\mathcal{S} = \{i_1, \dots, i_R\}$, where $R = |\mathcal{S}|$. Their samples are concatenated into a one-dimensional signal

$$u_c[n] = x_{i_r, c, t}, \quad n = (r-1)T + t, \quad r \in \{1, \dots, R\}, \quad t \in \{1, \dots, T\}.$$

Thus $u_c \in \mathbb{R}^{RT}$ is the channel-specific signal on which blink onsets and offsets are estimated. If the analysis is run in the fallback mode described above, $\mathcal{S}$ is the full valid epoch set; otherwise it is the set of epochs identified as suspicious in Stage A.

Candidate blink samples are defined by a threshold-crossing indicator

$$h_c[n] = \mathbf{1}\{u_c[n] > \theta_c\}, \quad n = 1, \dots, RT.$$

After any required polarity normalization, $h_c[n] = 1$ indicates that the sample amplitude exceeds the channel-specific blink-region threshold. Blink regions are then defined as maximal contiguous runs of supra-threshold samples. A candidate interval for channel c is written as the half-open index range $[s_{c,r}, e_{c,r})$, where

$$h_c[n] = 1 \quad \text{for all } n = s_{c,r}, \dots, e_{c,r} - 1,$$

with either $s_{c,r} = 1$ or $h_c[s_{c,r} - 1] = 0$, and either $e_{c,r} = RT + 1$ or $h_c[e_{c,r}] = 0$. Equivalently, the onset $s_{c,r}$ is the first sample in a run that crosses above θ_c , and the offset $e_{c,r}$ is the first subsequent sample at which the signal has returned to the sub-threshold state.

To suppress brief threshold crossings that are unlikely to represent complete blinks, each candidate interval is required to exceed a minimum duration. Let f_s be the sampling frequency and let $\ell_{\min}$ be the minimum blink duration in seconds. The corresponding sample length is

$$L_{\min} = \ell_{\min} f_s.$$

The accepted blink-region set for channel c is

$$\mathcal{R}_c = \{(s_{c,r}, e_{c,r}) : e_{c,r} - s_{c,r} > L_{\min}\}.$$

This duration gate is applied independently for each channel. Consecutive supra-threshold samples are merged into the same region by construction, while two runs separated by at least one sub-threshold sample are treated as distinct candidate blink regions.

Finally, the accepted intervals are mapped from concatenated coordinates back to epoch-local coordinates using the known epoch boundaries. If an accepted onset $s_{c,r}$ lies in the concatenated segment corresponding to epoch i_q , its within-epoch onset sample is

$$\hat{s}_{c,r} = s_{c,r} - (q-1)T,$$

and its onset time is $(\hat{s}_{c,r} - 1)/f_s$ seconds relative to the beginning of that epoch. The interval duration is

$$\hat{d}_{c,r} = \frac{e_{c,r} - s_{c,r}}{f_s}.$$

The final output of this stage is therefore a channel-indexed collection of blink regions, each represented by an epoch identifier, a channel label, an onset index or time, and an offset or duration. The extraction remains channel-wise throughout: the threshold θ_c estimated for channel c is applied only to that channel's signal, and the resulting regions $\mathcal{R}_c$ are retained as channel-specific blink candidates.

3.4 Problem Definition

Let a blink annotation be an interval $g_i = [s_i, e_i]$ and a detector output be an interval $d_j = [\hat{s}_j, \hat{e}_j]$. A detected region is counted as a true positive if it overlaps at least one unmatched ground-truth region. Unmatched detections are false positives; unmatched ground-truth regions are false negatives. This event-level treatment follows the logic of interval-based detection rather than pointwise accuracy, which is appropriate because blink boundaries are uncertain, annotation granularity varies, and non-event samples dominate the timeline (Saito and Rehmsmeier, 2015; Salles et al., 2024).

3.5 Datasets

We validated our methods on four open datasets with data from over 200 subjects. This allowed us to evaluate experimental strengths and potential limitations of different rejection methods. The datasets contained either EEG or MEG data. To obtain solid experimental conclusions, diverse experimental paradigms were considered with data from working memory, perceptual and motor tasks.

The primary naturalistic dataset is the sustained-attention driving corpus of Z. Cao et al. (2019), which contains 62 sessions of 32-channel EEG recorded during a 90-minute lane-keeping task. Its official event schema includes deviation onset, response onset, and response offset, and the published preprocessed version explicitly notes manual removal of apparent eye blinks. Therefore, this corpus is suitable for the present study only in its raw form and only when paired with externally available blink-region annotations.

The second dataset named in the project brief requires correction. PMID 30325349 refers to the Scientific Data motor-imagery corpus of Kaya et al. (2018), not to a blink-annotated benchmark. It is therefore inappropriate as a primary supervised blink dataset. In the present design it is repurposed as an auxiliary robustness corpus for false-positive stress testing: because it contains frontal EEG but lacks blink-event labels, it can help measure how often a threshold family spuriously fires in out-of-domain EEG.

Because the paper aims to be scientifically robust rather than merely brief-compliant, we recommend adding the 2025 multimodal ocular-activity dataset by Guttmann-Flury, Sheng, and Zhu (2025) as a companion benchmark. Its synchronized EEG, eye tracking, and high-speed video make it the strongest official source for validating blink-region decisions and studying boundary uncertainty.

3.6 Preprocessing

All threshold families are evaluated under the same preprocessing backbone. We recommend using frontal or prefrontal channels only, after polarity normalization if needed, detrending, and a blink-oriented band-pass in the low-frequency region. Candidate maxima are then detected from the preprocessed signal or a derived blink component, and candidate windows are expanded by local valley search or hysteresis rules. This standardization is necessary because the literature shows that filter choice, candidate signal choice, and reference scheme can strongly alter the apparent success of a threshold (Chang et al., 2016; Kleifges et al., 2017; Tran et al., 2021; Y. Zhang et al., 2023).

3.7 Threshold Families

3.8 Evaluation

The primary metrics are event-level precision, recall, and F1 under non-zero-overlap matching, with secondary analyses using temporally tolerant variants or IoU-weighted summaries. Precision,

recall, and F1 are preferred because blink events are sparse relative to non-blink samples; raw accuracy would be inflated by true negatives and would obscure detector behavior on the rare events of interest. Soft temporal metrics are reported secondarily because interval boundaries are often approximate and neighboring detections can still be operationally useful (Saito and Rehmsmeier, 2015; Salles et al., 2024).

Per-session estimates should be macro-averaged, with bootstrap confidence intervals across sessions and a paired nonparametric comparison of threshold families on session-level F1. For the motor-imagery corpus, the relevant endpoint is not F1 but false-positive burden per minute and per hour, since the corpus is used only for robustness stress testing.

4 Experiments and Results

4.1 Experimental Setup

All experiments are evaluated on the driving corpus described in Section 3.5. Unless otherwise stated, each continuous session is divided into non-overlapping 30-second epochs (selected by Experiment 1), and the evaluation is conducted at the event level using the overlap-based matching criterion introduced in Section 3.4.

Conditions. Five conditions are compared throughout the main experiments.

BLINKER-concat. The BLINKER algorithm is applied to the full concatenation of all valid epochs, treating the session as a single continuous signal and deriving a single threshold from the entire recording (Kleifges et al., 2017). This condition represents the current practice of running a continuous-signal detector on epoch-structured data without any epoch-aware adaptation.

DBO. Direct Bayesian Optimisation of the sample-level amplitude threshold, applied to the concatenated signal without any preceding epoch-screening stage. This ablation isolates the value added by Stage A: if DBO alone recovers competitive performance, the epoch-screening step is unnecessary.

MNE-annot. The MNE-Python `annotate_amplitude` routine, which marks contiguous segments whose peak-to-peak amplitude exceeds a configurable threshold, used here with its default parameterisation. This baseline represents widespread preprocessing practice in the neuroimaging community and provides a reference point grounded in community tooling.

Proposed-Mean. The full three-stage pipeline described in Section 3, with the sample-level threshold at Stage B estimated from the arithmetic mean and standard deviation of samples pooled from suspicious epochs.

Proposed-Med. The three-stage pipeline with the sample-level threshold at Stage B estimated from the median and median absolute deviation (MAD). This is the primary proposed configuration.

4.2 Evaluation Metrics

Event-level precision, recall, and F1. A detected interval is counted as a true positive if it overlaps at least one unmatched ground-truth blink region; unmatched detections are false positives, and unmatched ground-truth regions are false negatives. Per-session precision, recall, and F1 are computed from these event-level counts.

Macro versus micro F1. We adopt **macro-averaged F1** as the primary reporting metric. Macro F1 computes precision, recall, and F1 independently for each session and then reports the unweighted mean of the session-level F1 scores. This gives every session equal weight regardless of how many blink events it contains. The goal of epoch-based blink detection in driving studies is to produce a detector that works reliably session by session, not one that merely accumulates favourable aggregate counts. Micro F1, which pools all true positives, false positives, and false negatives across sessions before computing a single ratio, is dominated by sessions with large blink counts and can mask consistent failure modes on sessions with atypically few blinks. Both metrics are reported, but all hypothesis tests are conducted on the vector of session-level F1 scores, for which each session contributes exactly one observation.

Wilcoxon signed-rank test. Pairwise comparisons between conditions use the Wilcoxon signed-rank test on matched pairs of session-level F1 scores. The test is preferred over a paired t -test because session-level F1 distributions are bounded and often skewed. For comparisons in which the proposed method is hypothesised to outperform a baseline, a one-tailed test is used; stability analyses use two-tailed tests. Bonferroni correction is applied for the number of conditions compared within each experiment, and we report exact p -values alongside the matched-pairs rank-biserial correlation r as an effect-size estimate.

4.3 Experiment 1: Compare different Strategy

4.3.1 Strategy A: Naive Epoch Concatenation as a Proxy for Existing Practice

Motivation. In driving and other epoch-structured paradigms, it is common to run a continuous-signal blink detector on the raw or minimally preprocessed recording and subsequently map detections back to the epoch grid. BLINKER-concat embodies this practice: its threshold is derived from a signal pool that mixes blink-heavy and blink-free epochs. The central claim of the proposed pipeline is that this mixing dilutes the amplitude statistics used for threshold estimation, pushing the threshold upward and reducing recall. Experiment 1 tests this claim directly.

Hypothesis. Naive concatenation will produce systematically lower session-level F1 than the proposed epoch-aware pipeline, primarily through reduced recall rather than reduced precision.

Design. All five conditions are applied to every session under 30-second epochs. Per-session macro F1, precision, and recall are summarised in Table 2.

4.3.2 Strategy B: MNE-Based Annotation as a Community Baseline

Motivation. MNE-Python is the most widely used open-source EEG analysis platform in neuroscience. Its `annotate_amplitude` routine provides a threshold-based artifact marking mechanism that many practitioners reach for when a specialised blink detector is unavailable. Establishing its performance under an event-level blink criterion, for which it was not designed, characterises the gap that specialised methods must close.

Design. The MNE-annot condition is drawn from Table 2. As a secondary diagnostic, the detection timestamps produced by MNE-annot are aligned to those from Proposed-Med, and the distribution of temporal offsets at onset and offset boundaries is reported. Systematic offsets indicate whether MNE-annot consistently detects the right events but with shifted boundaries, or whether the discrepancies arise from entirely different detection triggers.

4.3.3 Strategy C: Direct Bayesian Optimisation Without Epoch Screening

Motivation. Bayesian optimisation (BO) has shown promise for threshold tuning in related biosignal detection problems. A natural question is whether applying BO directly to the concatenated signal, bypassing Stage A entirely, is sufficient to achieve competitive performance. The DBO condition answers this question.

Hypothesis. DBO will outperform BLINKER-concat—because BO can adapt the threshold to each session—but will underperform Proposed-Med. The expected shortfall arises because DBO optimises the threshold over a distribution that mixes blink-containing and blink-free signal segments, whereas Stage A concentrates Stage B estimation on segments enriched for blink-like excursions. The magnitude of the gap should scale with the fraction of clean epochs in the session.

Diagnostic. For each session we record the fraction of epochs identified as suspicious by Stage A, and plot this fraction against the session-level F1 gap between DBO and Proposed-Med. A positive association would confirm that Stage A’s filtering benefit scales with the proportion of clean epochs in the session—the more clean epochs, the more dilution DBO suffers, and the more beneficial the epoch-screening stage becomes.

4.3.4 Strategy F

4.4 Experiment 2: Effect of Threshold Estimator at Stage B

Motivation. For the strategy F, under it its Stage B, the Stage B estimates the sample-level threshold from samples pooled across the suspicious epoch set $\mathcal{S}$. Because $\mathcal{S}$ is selected precisely because it contains high-amplitude excursions, the pooled distribution is not symmetric around a clean baseline: it is contaminated by the very peaks the threshold is meant to capture. The mean and standard deviation are sensitive to these extreme values; the median and MAD are not.

Hypothesis. Proposed-Med will outperform Proposed-Mean on precision—because the MAD-based threshold is less inflated by within-epoch peaks—while achieving comparable or superior recall, leading to a higher F1. The benefit of the robust estimator is expected to be larger for sessions whose suspicious-epoch pools contain extreme outlier amplitudes.

Design. Proposed-Mean and Proposed-Med are compared using the Wilcoxon signed-rank test on session-level F1 (Table 2). A supplementary scatter plot shows the per-session F1 difference against the 95th-percentile amplitude in the suspicious-epoch pool, used here as a proxy for outlier severity.

Instruction Agent We can run this together with the Experiment 1, but the strategy F should run first the mean, and second the median.

4.5 Experiment 3: Stability Across Epoch Durations

Motivation. Epoch duration is routinely chosen on paradigm-specific or theoretical grounds and is rarely optimised for the blink detector itself. In driving studies, epoch lengths ranging from 30 seconds to several minutes appear in the literature. A robust pipeline should produce stable event-level F1 across this range, because the underlying physiology of blinks does not change with the epoch grid imposed during analysis.

Design. Proposed-Med is evaluated under epoch durations of 20, 30, 40, 60, and 30 seconds. For each duration the full pipeline is re-run from scratch, since epoch-level peak-to-peak statistics change with epoch length. Primary outcomes are per-session macro F1; secondary outcomes are the number of suspicious epochs returned by Stage A and the estimated sample-level threshold θ_c . Table 3 reports results and two-tailed Wilcoxon p -values relative to the 60-second reference.

Instruction Agent This is only for the strategy F, but we evaluate using different epoch size

4.6 Experiment 4: Stability Across Peak-Side Boundary Tolerance

Motivation. Event-level matching requires specifying how much temporal mismatch between a detected interval and a ground-truth interval is tolerable. Blink boundaries in annotation are inherently uncertain: different annotators and annotation tools locate onset and offset slightly differently. A detector whose F1 score is highly sensitive to the matching tolerance is fragile in practice, because its apparent performance depends on an arbitrary methodological choice rather than on true detection quality.

Design. Proposed-Med is evaluated under five event-matching strictness settings, expressed as intersection-over-union (IoU) thresholds $\{0.0, 0.1, 0.2, 0.3, 0.5\}$ for the overlap between each detected interval and a ground-truth interval. Lower IoU thresholds treat any overlap as a match; higher thresholds require a larger fraction of overlap and therefore penalise boundary mismatch more aggressively. We report macro-averaged precision, recall, and F1 at each IoU threshold, and the range of macro F1 across the five levels to quantify sensitivity to this evaluation choice.

4.7 Experiment 5: Robustness Under User-Initiated Pre-Rejection

Motivation. In operational driving-study pipelines, an analyst routinely discards epochs on quality grounds—gross movement artefacts, electrode pop-offs, or signal saturation—before any automated processing begins. To mimic this real-world scenario, we simulate user-initiated pre-rejection by withholding a random subset of valid epochs from the pipeline’s input. These pre-rejected epochs are removed from $\mathcal{E}$ and never enter Stage A. The pipeline must therefore tolerate a reduced and potentially non-representative input set, which is a distinct challenge from the epoch-screening performed internally by Stage A.

Design. For each session, a random subset of valid epochs is withheld before Stage A at withholding rates of 0%, 20%, and 40%. The full pipeline is then run on the remaining valid epochs. Per-session macro F1 is recorded; each withholding rate is replicated over 20 independent random draws of the withheld set, and session-level F1 is averaged across replications before the Wilcoxon test. Two-tailed Wilcoxon tests compare each non-zero withholding rate against the zero-withholding reference.

This experiment establishes that the pipeline handles the realistic case in which the analyst has already removed some epochs before invoking the blink detector, and characterises the withholding rate at which pre-rejection begins to harm performance.

4.8 Experiment 6: Morphological Analysis of Missed and Spurious Blinks

Motivation. Aggregate F1 scores report whether the detector is accurate overall, but they do not reveal which blink morphologies the algorithm consistently misses or misidentifies. False negatives

may cluster at short blinks, at blinks with atypical waveform shapes, or at blinks co-occurring with saccadic activity. Identifying these failure modes guides future algorithmic refinement.

Design. For the Proposed-Med condition, we extract a 500-millisecond waveform window centred on the peak of each ground-truth blink and categorise it as true positive (TP), false negative (FN), or the nearest unmatched detected peak as false positive (FP). Butterfly plots—overlaid single-trial waveforms from the frontal channel with the highest mean blink amplitude—are constructed separately for TP, FN, and FP events.

If false-negative blinks are morphologically distinct—for instance, consistently lower peak amplitude, compressed rise-time, or co-occurring saccadic deflections in adjacent channels—those features will be visually salient in the FN butterfly relative to the TP butterfly. FP butterflies similarly reveal whether spurious detections arise from saccades, slow drifts, or muscle transients, pointing to specific thresholding assumptions that may need revision.

4.9 Experiment 7: Sensitivity to the Minimum Suspicious-Epoch Count $n_{\min}$

Motivation. Stage B falls back from $\mathcal{S} = \mathcal{B}$ to $\mathcal{S} = \mathcal{E}$ when the number of suspicious epochs is smaller than $n_{\min}$. Setting $n_{\min}$ too low risks deriving a threshold from a handful of epochs that may not represent the full range of blink amplitudes in the session; setting it too high triggers the fallback unnecessarily and discards the epoch-filtering benefit of Stage A. The appropriate value of $n_{\min}$ is currently left as a design choice, and this experiment provides empirical guidance.

Threshold-variance analysis. For sessions in which Stage A returns at least 20 suspicious epochs, we randomly subsample $\mathcal{B}$ down to $n \in \{2, 3, 5, 7, 10, 15, 20\}$ epochs and recompute the Stage B threshold θ_c for 50 independent subsamples at each level. The inter-subsample standard deviation of θ_c is plotted against n . The smallest n at which this variance stabilises—identified as a knee point in the curve—provides an empirical lower bound for $n_{\min}$.

Fallback-frequency analysis. For each epoch duration in Experiment 5, and for candidate values $n_{\min} \in \{3, 5, 10\}$, we compute the fraction of sessions for which $|\mathcal{B}| < n_{\min}$ and the fallback therefore triggers. Shorter epochs increase the absolute epoch count, which may raise $|\mathcal{B}|$ even when the contamination fraction is constant. This analysis therefore reveals whether the choice of $n_{\min}$ interacts with epoch duration, and which combinations of duration and $n_{\min}$ lead to excessive fallback.

Together, the two analyses yield a principled recommendation: use the smallest $n_{\min}$ at which threshold-variance stabilises, and verify that this value does not trigger the fallback for more than a small fraction of sessions under the chosen epoch duration.

4.10 Experiment 8: Data-Driven Selection of the DBO Scan-Scale Hyperparameter

Motivation. Strategy C (DBO) uses Bayesian optimisation to learn per-channel amplitude thresholds from the EEG data. These *raw thresholds* are then multiplied by a fixed scale factor, `STAGE1_SCAN_SCALE`, to produce the *scanning threshold* used for blink detection:

$$\theta_{\text{scan}}^{(c)} = \text{STAGE1_SCAN_SCALE} \times \theta_{\text{raw}}^{(c)}.$$

The current default of 0.12 was chosen manually. As noted in Section 4.3.3, detection performance is sensitive to this choice: too small a scale gives a very permissive threshold that admits spurious

detections (low precision); too large a scale misses low-amplitude blinks (low recall). A data-driven selection of the optimal scale is therefore warranted.

Design: Gaussian-process Bayesian optimisation. We search for the scan scale in log space over $[0.01, 2.00]$ using Gaussian-process Bayesian optimisation. The run uses 25 total scale evaluations: five random initial evaluations followed by 20 expected-improvement-guided evaluations. For each Raja session, the expensive Bayesian-optimisation-based autoreject thresholds are computed once at scale 1.0, and each candidate `STAGE1_SCAN_SCALE` is then evaluated as a cheap multiplicative post-step on the precomputed raw thresholds. The objective is macro-averaged F1 across the 46 Raja sessions.

Results. Table 4 reports the observed Raja macro-F1 values for the unique evaluated scales. The best observed value was `STAGE1_SCAN_SCALE` = 0.1388, with macro-F1 = 0.8702. This is close to, but measurably higher than, the previous manual default of 0.12; we therefore recommend replacing the default with 0.1388 for the DBO configuration evaluated here. The independent Cao-2018 validation stage was initiated but not completed in this run, so it is not reported in this table.

5 Discussion

Epoch duration selection. Experiment 1 selected a 30-second epoch duration for the Proposed-Med configuration using macro-F1 on the combined dataset as the primary metric. Importantly, Experiment 3 showed that macro-F1 was largely insensitive to the epoch grid over 20–120 s, with only small differences between durations and no evidence of a strong performance cliff. This supports the practical view that epoch duration is a secondary design choice for the detector itself; we nevertheless fix 30 s downstream for consistency and to match the empirically best setting under the chosen metric.

Strategy comparison. Across the five compared conditions, Proposed-Med (macro-F1 = 0.8177) did not dominate DBO (macro-F1 = 0.8540) under the selected 30-second epoch. Stage A may remove epochs with atypical but valid blink morphologies, trading recall for robustness. Future work should quantify when Stage A is beneficial and when it is overly conservative. Proposed-Med substantially improved over the two practical baselines (BLINKER-concat, macro-F1 = 0.7453; MNE-annot, macro-F1 = 0.5610).

Sensitivity to the evaluation strictness. Experiment 4 demonstrated that macro-F1 varies by 0.5168 across IoU thresholds $\{0.0, 0.1, 0.2, 0.3, 0.5\}$ (stable <1 pp: NO). The main experiments therefore adopt IoU = 0.1 as a pragmatic default, while reporting the full IoU sweep to make this methodological dependence explicit.

Morphological analysis. Experiment 6 (detailed) complements the scalar metrics by visualising typical true positives, false positives, and false negatives across duration and amplitude strata. This helps distinguish failures due to boundary misalignment from failures due to missed events, and provides qualitative evidence about which blink types are systematically challenging.

6 Conclusion

This work frames blink detection in epoch-structured EEG as both a detection problem and a threshold-selection problem. Using a shared event-level evaluation protocol, we compared five strategies and performed targeted stability analyses to understand how methodological choices (epoch length and event-matching strictness) affect reported performance.

Under the primary Proposed-Med configuration, a 30-second epoch duration achieved the best macro-F1 among the tested durations (20, 30, 40, 60, 120 s), with performance remaining stable across this range (macro-F1 = 0.8177 at 30 s). Strategy comparisons showed that the proposed epoch-aware pipeline substantially improves over naive concatenation and a community baseline (MNE `annotate_amplitude`), while a direct Bayesian-optimisation ablation can remain competitive in macro-F1. Finally, the IoU sweep highlights that evaluation strictness must be reported explicitly: stricter overlap thresholds lead to markedly lower F1 even when detections are near-misses.

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Table 1: Summary of experiments (Sec. 4.1–4.6). Experiments 1–7 test the robustness and correctness of the proposed three-stage pipeline across conditions, epoch durations, boundary tolerances, pre-rejection scenarios, event morphologies, and parameter sensitivity.

Script	Experiment and Description
40_exp1_exp2_strategy_comparison.py	Exp 1 & 2: Compares all five conditions (BLINKER-concat, MNE-annot, DBO, Proposed-Mean, Proposed-Med) on both Raja and murat2018 datasets at 60-second epochs. Proposed-Mean runs before Proposed-Med (as specified in experimental design). Includes per-session macro-F1 results, Wilcoxon signed-rank tests with Bonferroni correction for 10 pairwise comparisons (one-tailed for proposed-vs-baseline, two-tailed for Mean-vs-Med), and micro-/macro-averaged metrics per condition.
41_exp3_epoch_duration.py	Exp 3: Evaluates Proposed-Med (Strategy F with median estimator) across four epoch durations: 20, 30, 60, and 120 seconds on both Raja and murat2018 datasets. For each duration, re-runs the entire pipeline from scratch (Stage A–C) since epoch-level statistics change with epoch length. Reports per-session macro-F1, number of flagged epochs (n_{flagged}), and estimated threshold θ_c . Two-tailed Wilcoxon signed-rank tests compare each non-reference duration against the 60-second baseline (Bonferroni correction for 3 comparisons).
42_exp4_boundary_tolerance.py	Exp 4: Tests Proposed-Med stability across five boundary tolerances for event-level matching: 0, 50, 100, 150, and 200 milliseconds on both Raja and murat2018 datasets. Runs the pipeline once per session and evaluates against all tolerance levels to avoid redundant computation. Reports macro-averaged precision, recall, and F1 at each tolerance, and computes the range of macro-F1 across the five levels (range < 1 percentage point indicates stability).
43_exp5_user_rejection.py	Exp 5: Simulates user-initiated pre-rejection of epochs by withholding 0%, 20%, and 40% of valid epochs before Stage A. For each non-zero withholding rate, 20 independent random draws are performed; session-level F1 is averaged across the 20 replicates before Wilcoxon testing. Ground truth remains fixed (blinks in withheld epochs count as false negatives). Two-tailed Wilcoxon tests compare each non-zero rate against the 0% reference (Bonferroni correction for 2 comparisons).
44_exp6_morphological.py	Exp 6: Performs morphological analysis of detected and missed blinks on both Raja and murat2018 datasets by extracting 500-millisecond waveform windows centred on the peak of each blink event (both detected and ground-truth). Events are categorized as true positive (TP), false negative (FN), or false positive (FP). Waveforms are drawn from the frontal channel with the highest mean peak amplitude. Generates butterfly plots (overlaid single-trial waveforms with mean highlighted) for TP, FN, and FP categories per dataset, saved as PNG files for visual inspection of failure modes.
45_exp7_nmin_sensitivity.py	Exp 7: Two-part analysis of the fallback threshold n_{min} (minimum suspicious-epoch count for Stage B to use flagged epochs) on both Raja and murat2018 datasets. <i>Part A</i> (threshold-variance): For sessions with ≥ 20 flagged epochs, randomly subsamples the flagged set to $n \in \{2, 3, 5, 7, 10, 15, 20\}$; for each n , computes Stage B threshold over 50 independent subsamples and records the inter-subsample standard deviation. The smallest n at which variance stabilises provides an empirical lower bound for n_{min} . <i>Part B</i> (fallback-frequency): For each epoch duration in 20, 30, 60, 120 s and each candidate $n_{\text{min}} \in \{3, 5, 10\}$, reports the fraction of sessions triggering fallback ($ \mathcal{B} < n_{\text{min}}$).

Table 2: Main comparison on 30-second epochs. Values are macro-averaged across sessions (mean $\pm$ SD). Best value per column is shown in **bold**. † indicates $p < 0.05$ versus Proposed-Med (Wilcoxon signed-rank, one-tailed, Bonferroni-corrected for 10 comparisons).

Condition	Precision	Recall	F1
BLINKER-concat	0.6478 $\pm$ 0.2547	0.9693 $\pm$ 0.0382	0.7453 $\pm$ 0.2094 †
MNE-annot	0.7115 $\pm$ 0.2929	0.5604 $\pm$ 0.4075	0.5610 $\pm$ 0.3833 †
DBO	0.8266 $\pm$ 0.1787	0.9128 $\pm$ 0.1207	0.8540 $\pm$ 0.1425
Proposed-Mean	0.8914 $\pm$ 0.1336	0.7771 $\pm$ 0.2472	0.8022 $\pm$ 0.2054 †
Proposed-Med	0.8875 $\pm$ 0.1337	0.8009 $\pm$ 0.2346	0.8177 $\pm$ 0.1946

Table 3: Macro F1 of Proposed-Med across epoch durations. p -values (Wilcoxon, two-tailed) compare each duration against the 60-second reference. A non-significant result indicates that performance is stable under that duration change. **Bold** row = best epoch selected for downstream experiments.

Epoch duration	Macro F1	p vs. 60 s
20 s	0.8160	0.4387
30 s	0.8177	0.7270
40 s	0.8176	0.4460
60 s	0.8112	(reference)
120 s	0.8133	0.1415

Table 4: DBO scan-scale tuning on the Raja corpus using GP Bayesian optimisation. Values are macro-averaged F1 across 46 Raja sessions at 30-second epochs. The selected scale was 0.1388.

BO phase	Scan scale	Macro-F1
Random	0.6038	0.0606
Random	0.1023	0.8499
Random	0.9455	0.0066
Random	0.4024	0.3532
Random	0.0165	0.4815
Guided	0.1495	0.8674
Guided	0.0507	0.7067
Guided	0.1289	0.8693
Guided	0.0100	0.4284
Guided	0.1898	0.8408
Guided	2.0000	0.0011
Guided	0.0762	0.7983
Guided	0.0305	0.5872
Guided	0.2372	0.7754
Guided	0.1172	0.8634
Guided	0.1654	0.8591
Guided	0.0630	0.7580
Guided	0.1388	0.8702
Guided	0.0226	0.5300
Guided	0.0899	0.8288